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Research Internship Report

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1. Introduction:

[1] To combat infectious diseases, vaccines are an essential part of public health. For proper antigen transport and greater activation of immunization, it is important to deliver vaccines in a precise way into the body. Vaccines are also incorporated with immunomodulatory adjuvants such as DNA, recombinant subunits, synthetic peptides, and nanoparticles to promote and sustain antigen-specific immune response in the body. To enhance the efficacy of the vaccine, reliable delivery systems are being synthesized that can target the specific anatomical sites and release the antigens gradually. Approaches such as polymer-based, nanoparticle-based based and three-dimensional scaffold-based structures for a delivery system have been made.

Among the biomaterials, Hydrogels, which are polymer-based delivery systems, have recently gained interest in targeted vaccine delivery since it is biocompatible, can interact with biological systems, and can modulate immune response. Moreover, hydrogels naturally act as adjuvants and amplify the immune response by releasing immunostimulatory agents in a controlled way. The hydrogels also upregulate pattern recognition receptors that can activate innate antiviral immunity. Due to their high water content, porosity, and elasticity, hydrogels can effectively interact with immune cells and present the antigens. Hydrogels hold the combination of both targeted delivery and adjuvant properties, which make them potential vaccine delivery systems. Delivery systems made from hydrogel take routes such as intranasal, transdermal, and oral to deliver vaccines to the immune cells.

1.2 Stimuli-responsive hydrogels for vaccine delivery:

[2]Hydrogels are a three-dimensional network of cross-linked hydrophilic polymers that can retain a large volume of water or biological fluids. Due to their high water content, they are regarded as biocompatible and resemble natural living tissue. Hydrogels can be formed through chemical, physical, or a combination of both chemical and physical crosslinking. The polymers can be natural, synthetic, or a combination of both. Hydrogels can perform volume phase transition or sol-gel transition in response to physiological stimuli, such as temperature, pH, ions, glucose, and external stimuli such as light and electrical signals. In this regard, they are known as ‘smart’ and intelligent carriers capable of releasing vaccines in a controlled and sustained way. [3] Since hydrogels are biocompatible, biodegradable, and non-toxic, they are being widely applied in biomedical applications for vaccine delivery. Their application ranges from wound dressing, tissue engineering, bio-sensing, bio-printing, to electrospinning.

[4,5] Conventional systemic delivery systems like bolus injections pose some issues, such as a lack of targetability, low immunogenicity, and unwanted inflammatory reactions. Localised delivery of vaccines to specific sites in the body has proven to be a more effective way of delivering than systemic delivery since it does not affect other normal tissues and delivers antigens to the target cells only. Therefore, stimuli-responsive hydrogels have gained interest as

vaccine carriers for localised therapy since they have high loading capacity and release antigens in a sustained and controlled way.

1.3. Thermoresponsive Hydrogels:

[6] Thermoresponsive hydrogels are a three-dimensional network of crosslinked polymer chains that can retain a large volume of water and release the vaccine in response to temperature changes in the body. The temperature changes determine the in situ gelling behavior of the hydrogel. Generally, aqueous solutions of thermoresponsive hydrogels are liquid at room temperature and gel at body temperature. To become gel at body temperature, the hydrogels need to undergo a phase transition, which is the sol-gel transition. Some polymer solutions become insoluble and form gels above the lower critical solution temperature, while others become insoluble and form gels below the upper critical solution temperature.

Different formulations of thermoresponsive hydrogels have been presented in different research articles.[7] In one article, using free radical polymerisation, a chemically crosslinked thermoresponsive hydrogel was synthesized using sucrose 1-O'-methacrylate (SMA), sucrose dimethacrylate, and N-isopropylacrylamide. The main findings are that the lower critical solution temperature, swelling properties, and the mechanical properties of the hydrogels can be controlled by changing the amount of SMA in the copolymer. Under pH 10 and body temperature, the hydrogels can undergo degradation due to the hydrolysis of the SMA ester bonds. The ability to adjust the thermoresponsiveness, degradability, and controlled drug release profile makes it a suitable vaccine delivery system. [8] In another article, a biodegradable, thermoresponsive hydrogel was synthesized by copolymerizing N-isopropylacrylamide with dextran macromer. The lower critical solution temperature of the hydrogel was found at 32°C. It swelled below the LCST and collapsed above this temperature. The drug release profiles showed that smaller molecules, such as methylene blue, were released faster at 37°C, and larger molecules, like proteins, were released in a sustained way at body temperature.

[9] To improve the mechanical strength, mucoadhesiveness, and create a sustained release profile of the hydrogel, casein, a natural protein, was added to thermoresponsive polymer Polaxamer 407. Due to its strong adhesiveness, it can remain in contact with the mucosa and also prevents the need for frequent administration of the vaccine, as it shows controlled and sustained release profiles.[10] Another study conducted on thermoresponsive polymers used PNPHO as a thermoresponsive polymer for intranasal delivery of protein or peptides. The hydrogel transformed into a gel in situ at physiological temperature. The hydrogel demonstrated very low toxicity and great encapsulation efficiency. Due to its fine targeting capabilities and sustained release profiles, PNPHO is considered a promising carrier for vaccine delivery.

[10] The dual redox and temperature-responsive hydrogel was synthesized based on copolymers of 2,2,6,6-tetramethylpiperidinyloxy methacrylate (TEMPO) and N-isopropylacrylamide

(NIPAM) in one study. The redox state of the TEMPO units (Hydrophobic radical vs hydrophilic oxoammonium radical) and the thermoresponsive nature of the NIPAM polymer in combination make the hydrogel dual responsive. The key findings from this study include the swelling behaviour of the hydrogel in response to both stimuli and also its ability to electrostatically encapsulate an anionic molecule. This study showed that the multi-responsive nature of the hydrogel makes it suitable for vaccine delivery, where reox triggers such as glutathione in biological environments and body temperature can be utilised for targeted release.

1.3. pH-Responsive Hydrogels:

[11,12] pH-responsive hydrogels are a three-dimensional network of crosslinked polymer chains that can swell or shrink in response to the pH of the environment. This is mainly due to the presence of carboxylic acids or amines on the polymer chains. In an acidic environment, basic groups such as amines become protonated by gaining a positive charge. Therefore, there is electrostatic repulsion between polymer chains, which allows the hydrogel to swell and absorb more water. In basic conditions, which have a high pH, deprotonation occurs in the acidic groups, and they gain a negative charge. This creates electrostatic repulsion between polymer chains and causes the network to swell. During the collapsing phase, the acidic and basic groups lose their charge and become neutral. In summary, pH-responsive hydrogels with carboxylic acid groups swell at high pH and collapse at low pH, whereas pH-responsive hydrogels with cationic groups swell at low pH and collapse at high pH. The changes in pH control the electrical charges in the polymer chains, which creates electrostatic repulsion and osmotic pressure in the 3D network of polymers. These physical forces help in swelling, which makes them suitable for the targeted delivery of vaccines in the body's different pH environments.

[13] In one study, researchers developed a pH-responsive hydrogel made of methacrylic acid and acrylamide for delivering silver nanoparticles in a controlled way to the infected wounds. It has been found that the hydrogel remains collapsed at acidic pH. In contrast, at alkaline pH, the hydrogel swells due to the ionization of the carboxylic acid groups, triggering the release of silver nanoparticles. This on-demand release mechanism makes it a potential stimuli-responsive vaccine carrier, and also its non-cytotoxicity to human cells makes it biocompatible. [14] In a recent study, a pH-responsive hydrogel was developed by grafting cellulose nanofibrils in a polyvinyl-alcohol borax network. Its responsiveness to pH changes was studied by the drug release studies, where the hydrogel released the drug 2.33 times greater at pH 5.4 compared to pH 7.4. This confirmed that the hydrogel will be suitable for releasing the drug in an acidic wound environment. For studying the morphology of the hydrogel, scanning electron microscopy was used. Fourier-transformed infrared spectroscopy was used for chemical structure analysis. This formulation of the pH-responsive hydrogel was found to be effective for antibacterial wound healing because of its biocompatibility, ability to act as an antioxidant, and controlled and sustained release profile in response to the pH changes.

[15] In a recent study, a dual pH-responsive bilayer hydrogel was developed by using a cationic polyacrylamide layer (PAAm) and an anionic polyacrylic acid layer (PAAC). The two layers had opposing swelling behaviour, where the polyacrylamide layer swells in basic pH and the polyacrylic acid layer swells in acidic pH. This dual response allows the hydrogel to swell across a wide pH range. Mechanical compression tests were conducted to measure the elastic modulus of each layer, to verify its bending response at both low and high pH. The ability to control the hydrogel's swelling operation at different pH gradients makes it suitable for targeted delivery.

[16] In another study, a pH as well as electro-responsive hydrogel was synthesized by polymerizing polyaniline with a bacterial cellulose matrix. At low pH, the bilayer hydrogel with a polyaniline layer was protonated and gained a positive charge, which repelled positively charged drugs and made it difficult to release the drug. At neutral and high pH, the polyaniline layer is negatively charged and hence allows and accelerates the release of the positively charged drug. Characterization methods, such as Fourier-Transformed Infrared Spectroscopy, verified the chemical structure of the hydrogel and the loading of the drug. Moreover, the hydrogel also responded to external electrical stimuli and enhanced the drug release kinetics of the hydrogel. Due to its biocompatibility and dual-responsive nature, this hydrogel system possesses high potential for targeted delivery of vaccines.

[17] An injectable pH-responsive hydrogel was developed by another research group to treat diabetics. The hydrogel was synthesized by covalently crosslinking N-carboxyethyl chitosan and adipic acid dihydrazide with hyaluronic acid. Its pH responsiveness behaviour is dependent upon the acylhydrazone bond in the chemical structure of the hydrogel. The chemical bond remains stable at neutral pH, and at low pH, it hydrolyses. The diabetic wounds have an acidic pH, which makes the hydrogel suitable for delivering the insulin. As usual, Fourier-transformed infrared spectroscopy was utilised to analyse and verify the chemical structure of the hydrogel. Scanning electron microscopy was used to analyze the porous structure of the hydrogel. For 14 days, the hydrogel provided controlled and sustained release of the insulin at the diabetic wound, reducing inflammation and promoting healing.

1.4. Limitations and Challenges:

Significant challenges are being faced in synthesizing stimuli-responsive hydrogels for viral vaccine delivery. One of the key limitations is the ability to shield the viral antigens from the acidic pH of the stomach and the enzymes present in the gastrointestinal tract during the oral vaccine delivery. The variation in gastrointestinal pH and enzymatic activities is the main challenge for the delivery of vaccines using the oral route. For the intranasal delivery route, the thermoresponsive hydrogels, such as those made from poloxamer, might undergo early sol-gel transition. For precision, thermoresponsive hydrogels must be in a sol (liquid) state at room temperature and gel at the nasal cavity temperature, which is around 34°C. So the primary limitation remains in achieving the sol-gel transition temperature with precision. In addition to

this, some hydrogels do not show a controlled and sustained release profile. Rather, they demonstrate a burst release profile after they enter the intestine.

[18,19] Sometimes, the initiators, organic solvents, or crosslinkers used in the formulations for the synthesis of the stimuli-responsive hydrogels might denature the viral vaccine structure inside the hydrogels. This can reduce the immunogenicity of the viral vaccine, leading to less effectiveness of its use. Often, many studies focus on just a single stimulus, such as pH or temperature. However, in reality, the body's internal environment is determined by different physiological conditions. Therefore, the dual-responsiveness nature of the hydrogels must be considered to get optimal results for the targeted, controlled, and sustained release of the vaccine.

[18] Furthermore, although hydrogels are biocompatible and demonstrate sustained and controlled release profiles, they can cause some complications in the delivery routes of the vaccine. For example, in the dermal route, the hydrogels might lead to dermatological issues, and then in the gastrointestinal tract, they might lead to gastrointestinal disturbances. Through the nasal and parenteral routes, they might lead to respiratory and cardiovascular complications. There is a chance of affecting the neural tissues if the hydrogels cross the blood-brain barrier.

[1] Another major challenge is to achieve optimal release kinetics of the viral antigens. If the release kinetics are similar to the conventional bolus injection, which is rapid, the delivery system's effectiveness might not be properly utilised. However, if the release kinetics are too slow and delayed, the immune activation might be compromised.

To summarise the limitations, the primary challenges for synthesizing vaccine delivery systems are:

- i) Proper Immune response activation: Antigen-mediated immune response must be boosted by the hydrogel-based delivery systems and enable a controlled and sustained release profile.
- ii) Enhanced Biocompatibility: To avoid inflammatory or unexpected immunogenic reactions, the hydrogels must, the hydrogel based systems must be biocompatible and maintain their structure until degradation.
- iii) Target-specific delivery: It is important to achieve controlled and site-specific delivery of the antigens to evoke both humoral and cell-mediated immunity. For long-term immunity, it is vital to activate memory B and T cells.

Therefore, it is crucial to experiment with different formulations of the hydrogels and perform different studies for the characterization of the hydrogel and analyze their effectiveness through understanding the results.

2. Objective and Aims:

The primary objective of the research project is to synthesize and characterise stimuli-responsive hydrogels for viral vaccine delivery through different synthesis processes, such as free radical polymerization and physical crosslinking by the cold method, and then compare which formulation would work best. Different characterisation techniques will be utilised to find the gelation temperature, swelling studies, porosity studies, and analysis of the chemical structure. The stimuli-responsive hydrogels will be synthesized to effectively encapsulate the viral antigens and protect them from the acidic environment of the stomach. The hydrogel-based delivery system will also release the viral vaccine to the target site under stimuli such as temperature and pH in a controlled and sustained way, without affecting the normal tissues.

To achieve the primary objective, the main aims of the research project are:

- i) To synthesize different formulations of pH and temperature-responsive hydrogels via different synthesis processes.
- ii) To characterise the performance of the pH-responsive hydrogels in vitro by imitating the gastrointestinal pH conditions, through swelling and stability studies.
- iii) To find the sol-gel transition temperature of the thermoresponsive hydrogel using different techniques according to the viscosity of the hydrogels.
- iv) To find viral antigen entrapment efficiency and release capabilities by analysing pore size, distribution, and chemical structure.
- v) To analyse the biocompatibility and safety of the hydrogels by performing cytotoxicity assays.

3. Materials and Methods:

In this section of the report, the materials and methods for the synthesis of four different formulations of stimuli-responsive hydrogels will be described. Three of the formulations are thermoresponsive, and the other is pH-responsive.

3.1. PAAc-Co-NHS Hydrogel:

The thermoresponsive hydrogel was synthesized using the oil-in-water emulsion polymerisation method, as shown in Figure 1. Two different formulations of the hydrogel were experimented and the gelation temperature of the formulations was determined using UV-VIS Spectrophotometry. Table 1 and 2 demonstrates the reagents, chemicals, and organic solvents used for each formulation.

Reagents	Amounts
Gelatin	1g
Acrylic Acid NHS Ester	0.1g
Acrylic Acid	3ml
Corn Oil	140ml
Tween	122ul
Water	20ml

Table 1: The amount of reagents and solvents used for formulation 1.

Formulation 1 was synthesized with Acrylic acid NHS Ester as the crosslinker, acrylic acid as the monomer, and gelatin as a backbone for the chemical structure. The corn oil was the continuous phase for the emulsion process, and the water was the aqueous phase. The surfactant used for this emulsion process was Tween.

With Formulation 1, an issue was encountered. The hydrogel did not gel at lower concentrations of the gel. So, for formulation 2, a two-factor experimental design was done to analyse which two factors were affecting the gelation of the hydrogel.

Reagents	Amounts
Gelatin	1g
Acrylic Acid NHS Ester	0.2g
Acrylic Acid	3ml
Corn Oil	20ml
Tween	122ul
Water	20ml

Table 2: The amount of reagents and solvents for formulation 2.

For formulation 2, the amount of the crosslinker Acrylic acid NHS Ester was increased from 0.1g to 0.2g, and the volume of corn oil was reduced from 140ml to 20ml to enhance the

emulsion process. The hydrogel was continuously washed in acetone and then in water and kept in the dessicator for drying.



Figure 1: Experimental Set-up for the synthesis method of oil-in-water emulsion polymerization.

The oil-in-water emulsion polymerization method was conducted in a three-necked round-bottomed flask at an initial temperature of 45°C, and stirred at 640 rpm. Later on, the reaction temperature was set at 55°C, at 1500 rpm. For formulation 1, the solidification of the hydrogel took more than 2 hours, so it was placed in the fridge overnight at 4°C. The oil-in-water emulsion polymerisation method includes dispersing oil droplets, which in this case is the corn oil, into a continuous aqueous phase, which is the water, and adding surfactant such as Tween, to stabilise the process.



Figure 2: The final image of the corn-oil hydrogel for formulation 2.

However, this formulation gelled at a very low temperature compared to the body temperature, and was a liquid state at high temperatures. Sodium Hydroxide was added dropwise to the hydrogel to increase the pH and neutralize excess acid present.

3.2. PAAc-Co-Dextrin Hydrogel:

The pH-responsive hydrogel was synthesized using the free-radical polymerisation method, as shown in Figure 3. For this hydrogel, two different formulations were also experimented with. The swelling studies were performed and two different pH and temperature conditions. Tables 3 and 4 demonstrate the reagents, chemicals, and organic solvents used for each formulation.

Reagents	Amounts
Dextrin	1g
Polyacrylic Acid	10ml
N, N'-Methylenebisacrylamide (MBAA)	0.08g
Water	85ml
Potassium Persulfate(KPS)	0.025g

Table 3: The amounts of reagents and solvents used for formulation 1.

The free-radical Polymerisation process was conducted in a three-necked round-bottomed flask. The monomers were dextrin and acrylic acid, while the crosslinker was MBAA and the initiator was KPS. The water was used as a solvent. First, the monomers and the crosslinkers were dissolved in 80ml of water, and then the KPS was dissolved in 5ml of water. The reaction temperature was kept at 70°C, and stirred at a rate of 400 rpm. The hydrogel was washed with acetone and then water for 4 four times and then kept in the dessicator overnight for drying.

With this formulation, the pH of the hydrogel was very low and hence, acidic. Therefore, the two-factor experimental design was utilised to find which two factors were affecting the low pH of the hydrogel, and formulation 2 was synthesized.

Reagents	Amounts
Dextrin	1g
Polyacrylic Acid	5 ml
N, N'-Methylenebisacrylamide (MBAA)	0.08g
Water	30 ml
Potassium Persulfate(KPS)	0.025g

Table 4: The amounts of reagents and solvents used for formulation 2.

For formulation 2, the volume of polyacrylic acid was reduced from 10ml to 5ml, and also the volume of water was reduced from 85ml to 30ml.

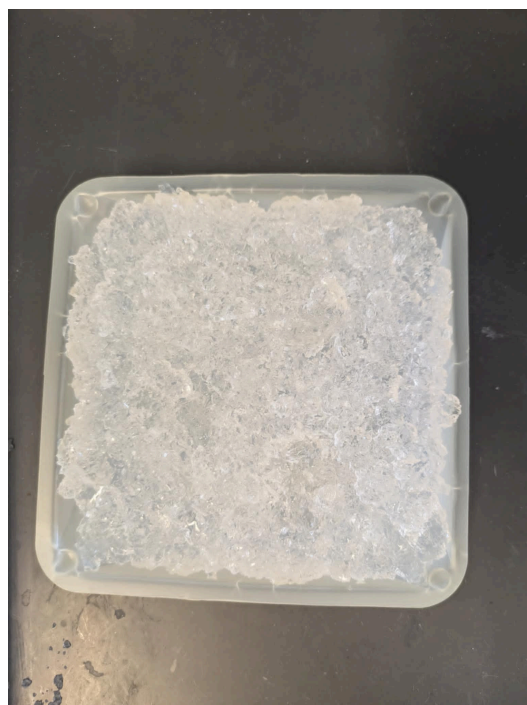


Figure 3: The final image of the Dextrin Hydrogel for formulation 2.

After the synthesis process, the pH of the hydrogel was adjusted to around 7.4 by adding sodium hydroxide dropwise. The sodium hydroxide neutralised the unreacted polyacrylic acid present in the hydrogel. After that, the hydrogel was kept in the dessicator overnight for drying.

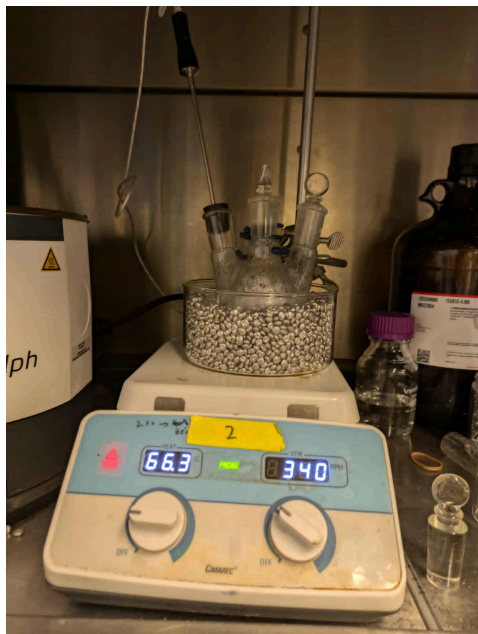


Figure 4: Experimental set-up for free-radical polymerisation technique.

3.3. PPLGA-Co-P407 Hydrogel:

This thermoresponsive hydrogel was synthesized by physical crosslinking using the cold method, as shown in Figure 5. For this hydrogel, only one formulation was experimented with. The amounts of reagents and solvents used are demonstrated in Table 5. The hydrogel was viscous, and so the gelation temperature was found using the tube-inversion method and a magnetic stirrer.

Reagents	Amounts
Pluronic F-127	6.6g
Pluronic F-68	0.6g
Hydroxypropyl Methylcellulose (HPMC)	0.3g
Water	30ml
Polyvinyl alcohol	0.28g
water	20ml
Poly lact-Glycolic Acid(PLGA)	0.2g

Table 5: Amount of reagents and solvents used for the experiment.

First, the PLGA microspheres were made one day before the synthesis of the thermoresponsive hydrogel. 0.2 g of PLGA was first dissolved in 1 mL of DCM, to create an organic phase, and 0.28g of PVA was dissolved in 20ml of water to create an aqueous phase. The organic phase was added dropwise to the aqueous phase, and then the emulsion was stirred for 3 hours. After that, it was centrifuged at 10,000 rpm to collect microspheres.



Figure 5: Experimental Setup for the physical crosslinking by the cold method synthesis process.

At 4°C, Pluronic F-127 and Pluronic F-68 were added to the chilled solution of the PLGA microspheres and were allowed to be stirred at 800 rpm using a magnetic stirrer. HPMC solution was added, and the reaction mixture was stirred continuously for 24 hours. To maintain the temperature at 4°C, ice packs and cubes were put around the three-necked round-bottomed flask, and later on kept in the fridge.

3.4. PHPMC-Co-P407 Hydrogel:

This thermoresponsive hydrogel is also synthesized by the cold method. Table 6 demonstrates the amount of reagents and solvents used to synthesize the hydrogel. The hydrogel solution was also viscous, so alternative methods to find gelation temperature, such as the tube-inversion method and magnetic stirrer, were utilised.

Reagents	Amounts
Hydroxypropyl methylcellulose (HPMC)	1.2g
Polaxamer 407	5.25g
Water	30ml

Table 6: Amounts of reagents used for the synthesis of the hydrogel.

First, the HPMC solution was prepared by adding 1.2g of HPMC to the 30ml of water in a beaker and stirring at room temperature until complete dispersion was achieved. Then, 5.25g of Polaxamer 407 was added gradually to the HPMC solution. The mixture in the beaker was transferred to a three-necked round-bottomed flask and stirred at 4°C for 4 hours at 350 r.p.m. Afterwards, the hydrogel solution was kept in the fridge for 24 hours at 4°C.

4. Results and Discussion:

In this section of the report, the results obtained from different studies to characterise the hydrogel will be demonstrated and analysed thoroughly. Different techniques to find gelation temperature, swelling coefficient, stability, chemical structure analysis, and cytotoxicity assays have been performed.

4.1 Gelation Studies:

i) PAAc-Co-NHS Hydrogel:

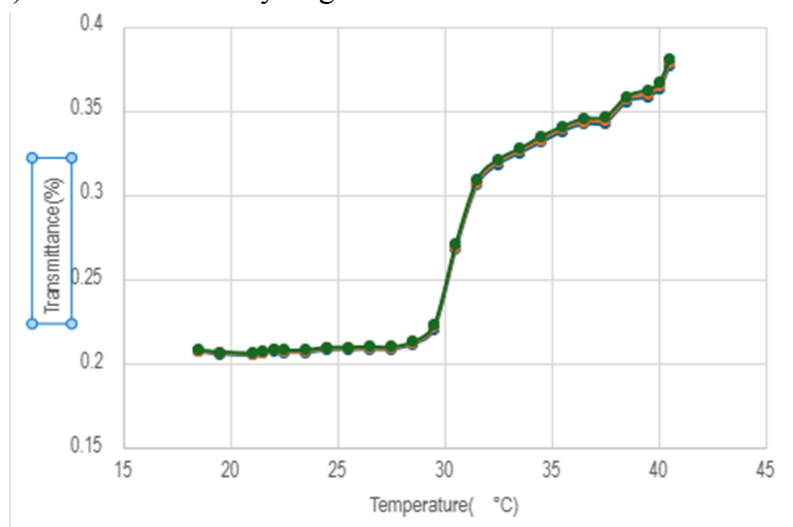


Figure 6: Transmittance over temperature curve obtained for formulation 1 after performing UV-VIS Spectrophotometry.

The PAAc-Co-NHS Hydrogel was not that viscous in consistency, so UV-VIS Spectrophotometry could be used to find the gelation temperature. At the temperature in the curve where there was the sharpest change in transmittance, that temperature was noted as the gelation temperature. This is because the sharp decrease in transmittance was due to an increased polymeric network in the hydrogel that occurred in the gel state. The gelation temperature was found to be 30.5°C. For formylation 2, the gelation temperature was found to be much lower than the body temperature, and the hydrogel was in a liquid state at high temperatures. This gelation temperature makes the hydrogel a suitable vaccine delivery system for the intranasal route since the temperature in the nasal cavity is around 25-30°C.

ii)PPLGA-Co-P407 Hydrogel: This hydrogel was very viscous, so an alternative method of finding gelation temperature using a magnetic stirrer and tube inversion method was performed.

Around 20ml of the hydrogel solution was put in a beaker and stirred with a magnetic stirrer at 50 r.p.m, as shown in Figure 7. The temperature at which the magnetic stirrer stopped rotating was determined as the gelation temperature.

Trial	Temperature(°C)
1.	32.400
2.	33.611
3.	31.303
4..	32.100

Table 7: Gelation temperature obtained at each trial.

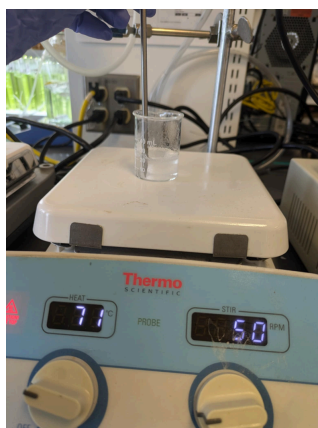


Figure 7: Setup for the gelation temperature study using a magnetic stirrer.

Tube-inversion method was also used to find the gelation temperature. For this, around 2ml of the hydrogel solution was put in a vial, and the vial was kept in a water bath. The temperature at

which the hydrogel started and stopped flowing was recorded. The gelation temperature was found to be the mean of the two temperatures.

Trial	Temperature (°C)
1.	29.362±3.01
2.	28.786±2.24
3.	27.885±2.52

Table 8: Gelation temperature results using the tube-inversion method.

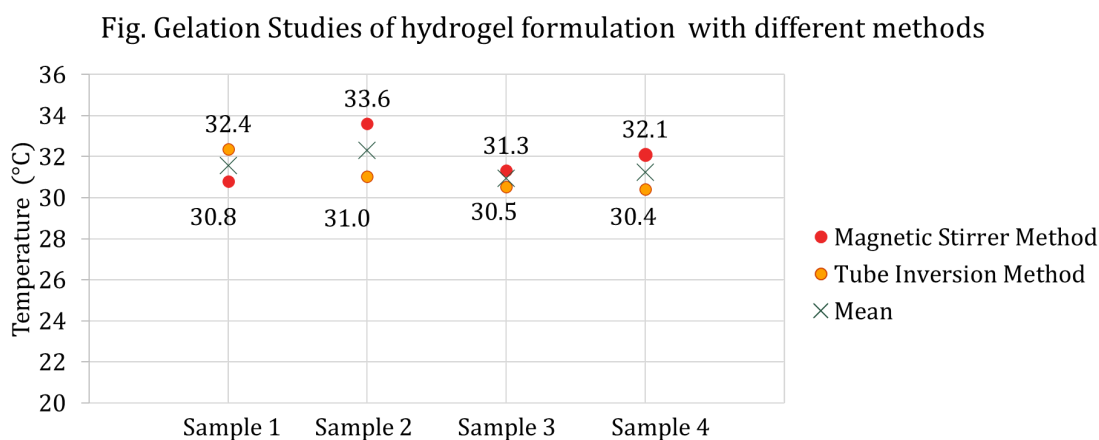


Figure 8: The gelation temperature of each sample using the two methods.

Overall, the mean gelation temperature lies between 30-34°C, making it suitable for intranasal delivery of a vaccine. According to the literature review, the gelation temperature is around 35-37°C. Therefore, our results are quite close to the literature findings.

iii) PHPMC-Co-P407 Hydrogel: This hydrogel was also quite viscous, so the tube-inversion method was utilised to find the gelation temperature.



Figure 9: Tube-inversion method procedure.

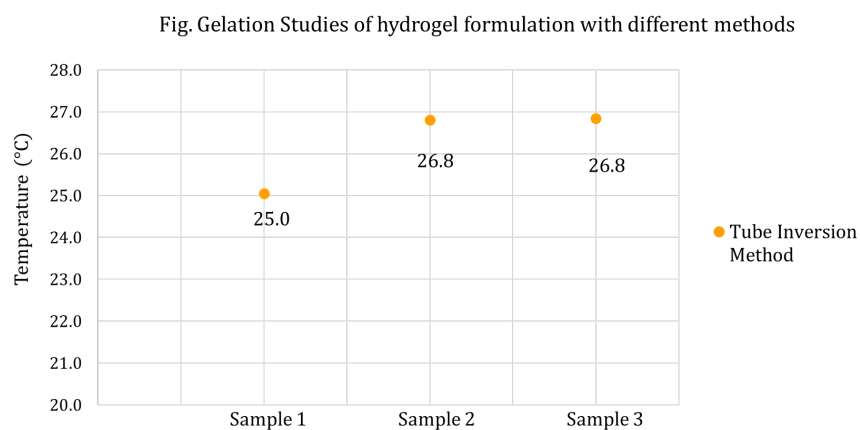


Figure 10: The gelation temperature results using the tube-inversion method.

The gelation temperatures, as shown in Figure 10, are between 25-26.8°C. In comparison to the gelation temperature found in the literature, which is around 37°C, it is lower, but this range is suitable for intranasal delivery of a vaccine. This can be due to the differences in the techniques used to find the gelation temperature.

4.2 Swelling Studies:

i) PAAc-Co-Dextrin Hydrogel

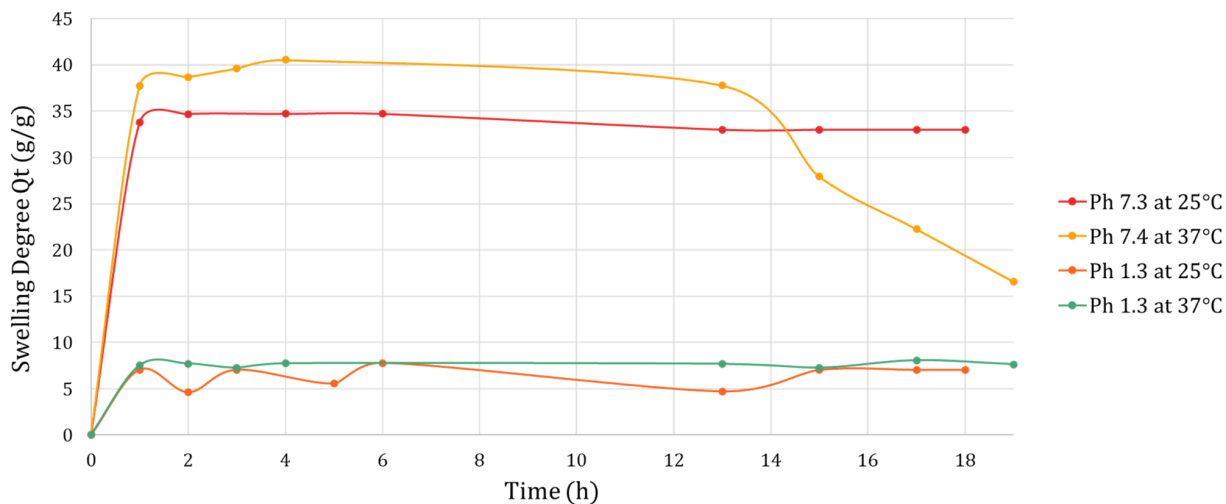


Figure 11: Swelling Degree over time graph for Dextrin hydrogel formula 2.

The swelling studies were conducted with the hydrogel at two different pH levels of 7.4 and 1.3, and two temperatures of 25°C and 37°C. The studies were conducted for two days. The initial mass was recorded for each sample used. After 1-hour intervals, the weight of the hydrogels was measured, and at each time, the swelling degree was found using the formula $Q_t = (W_f - W_0)/W_0$, where W_f is the weight of the sample at each time and W_0 is the initial weight.

The swelling curve in Figure 11 demonstrates that the hydrogel swelled more at pH 7.4 at each temperature. Moreover, the swelling degree was comparatively higher at 37°C at each pH. Therefore, as it swells, the pores open and the vaccine can diffuse out in the gastrointestinal cavity, which has a pH of 7.4, at a temperature of 37°C. At 25°C, the viral vaccine can be loaded in the hydrogel and be protected from diffusing out since it is in a collapsed state.

Fig. Swelling Formulation 1 Vs Formulation 2

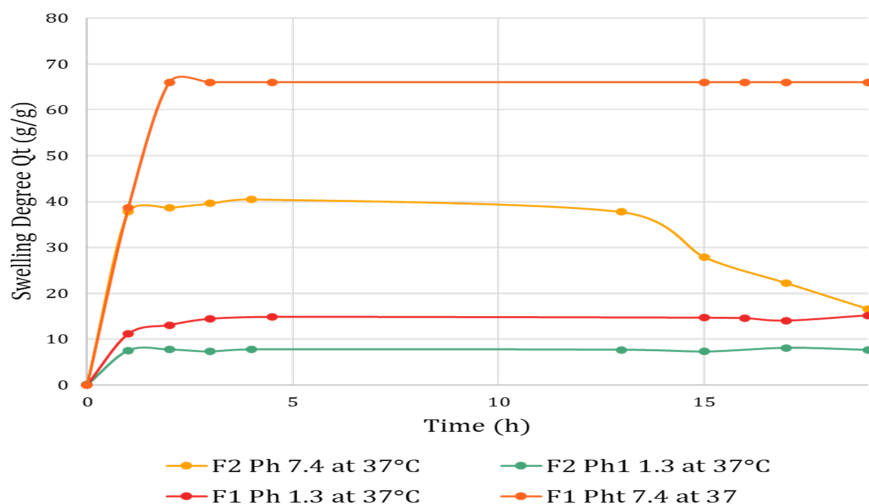


Figure 12: Comparison graph for the swelling behaviour of formulations 1 and 2.

Figure 12 shows that formulation 1 swelled more at pH 7.4 and 1.3 at a temperature of 37°C in comparison to formulation 2. This might be because formulation 1 had higher acid content than formulation 2, so more carboxylic acid groups and hence the polymer chains repelled as the carboxylic acid groups deprotonated at pH 7.4 and became negatively charged, allowing more swelling to occur.

4.3 Porosity Studies:

i) PAAc-Co-Dextrin Hydrogel

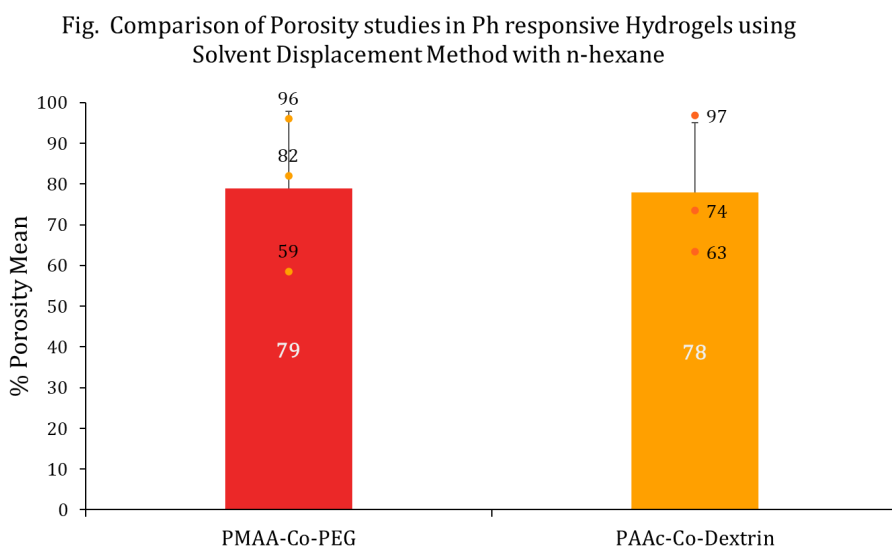


Figure 13: %Porosity Mean graph.

Porosity Studies were conducted using the solvent displacement method with n-hexane. It shows that the mean porosity is at 78% for the PAAc-Co-Dextrin hydrogel. This demonstrates that the hydrogel has a very good porous structure. The vast network of pores acts as a reservoir for the vaccine antigens to be loaded, and also demonstrates an efficient release profile.

4.4. Stability Studies:

i) PAAc-Co-Dextrin Hydrogel

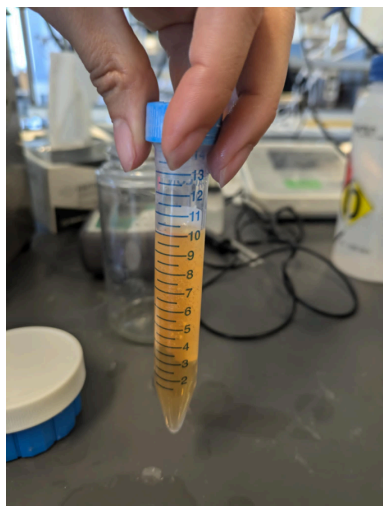


Figure 14: Dextrin Hydrogel in formulation 1 in DMEM media after 5 days.

The formulation 1 of the PAAc-Co-Dextrin Hydrogel was kept in DMEM media in a hot water bath at 37°C for 5 days. The color has changed to orange, which shows that the pH is still acidic.



Figure 15: The final mass of the dextrin hydrogel after 5 days.

The final mass was 2.673g, which was higher than the initial mass of 0.023g. This showed that the hydrogel swelled in the media at 37 °C. DMEM is a suitable cell culture medium and is suitable for stability studies as it simulates the in vivo environment of the body. The hydrogel did not degrade or dissolve completely, which shows that it is quite stable.

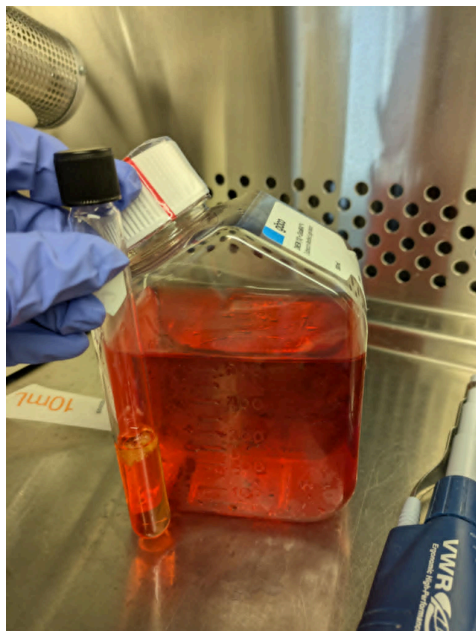


Figure 16: The dextrin hydrogel with formulation 2 in DMEM media.

The appearance of formulation 2 is quite the same as the DMEM media, which implies it is quite neutral. Similarly, it was kept at 37 °C in a dry water bath for 10 hours.



Figure 17: Final mass of formulation 2.

The final mass was found to be increased to 1.567 g from 0.023g, as the hydrogel swelled. This showed that the hydrogel was stable and did not degrade or dissolve in the media.

4.5. Dynamic Light Scattering Studies:

i)PPLGA-Co-P407 Hydrogel: Concentrations of 1, 0.5, and 0.25 mg/ml of hydrogel were made, and then dynamic light scattering was performed.

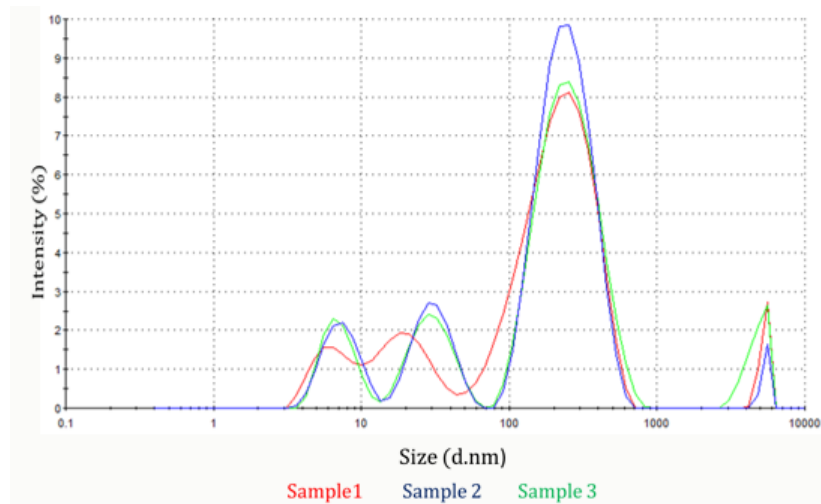


Figure 18: Size Distribution by intensity graph of the hydrogel.

Parameter	Value
Polydispersity Index (PDI)	0.583
z-average diameter (d.nm)	115.563

Table 9: The value of the polydispersity index and the average diameter of the particles.

The values of PDI and the average diameter of the particles can significantly indicate the size of the particles. The value of PDI is high for the hydrogel to act as a vaccine delivery system, and it shows that the hydrogel is highly polydisperse. The z-average diameter of the particles is within the range, so this ensures that the delivery system will have better tissue and cellular penetration, and targeted delivery.. The values can be enhanced by changing the purification steps.

ii)PHPMC-Co-P407 Hydrogel:

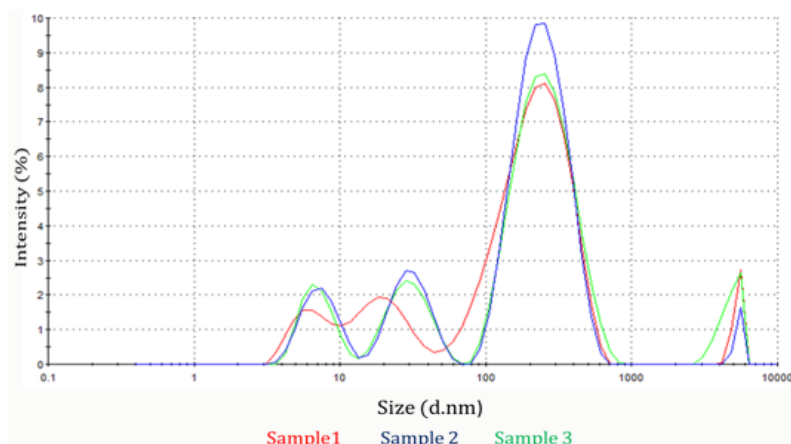


Figure 19: Size Distribution by intensity graph of the hydrogel.

Parameter	Value
Polydispersity Index (PDI)	0.583
z-average diameter (d.nm)	115.563

Table 10: The value of the polydispersity index and the average diameter of the particles.

Similarly, the value of PDI is high for the hydrogel to act as a vaccine delivery system, and it shows that the hydrogel is highly polydisperse. The z-average diameter of the particles is within the range, so this ensures that the delivery system will have better tissue and cellular penetration and targeted delivery. The values can be enhanced by changing the purification steps.

5. Conclusion:

In conclusion, this research project has successfully achieved its primary objective of synthesizing and characterizing stimuli-responsive hydrogels. Among the four formulations, PAAc-Co-Dextrin hydrogel and PPLGA-Co-P407 hydrogel demonstrated very good results for the swelling studies, stability studies, porosity studies, and gelation studies. With PHPMC-Co-P407 Hydrogel, the gelation temperature was also found, but the range is slightly lower than the expected temperature range. Still, it can work perfectly for intranasal delivery of vaccines. The PAAc-Co-NHS Hydrogel with formulation 2 showed unexpected gelation temperature results. However, with formulation 1, the gelation temperature could be found with UV-VIS Spectrophotometry. Furthermore, cytotoxicity assays were conducted by the Foothills Hospital team, and PHPMC-Co-P407 hydrogel demonstrated an average cytotoxicity range of 1-7%, which is quite good. Overall, the insights gained from these studies and tests enabled us to compare which formulations worked and the need to change the formulation or synthesis process.

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